PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Pentester Name (Cybersecurity Professional)	Shayan Rafiq
Program/Batch	Cybersecurity & Ethical Hacking Program – Networkwalks
Date	13 November 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM5 (Zenmap Scanning) W2-PM5 (Zenmap Scanning)
Client/Target	1. Networkwalks (secured written permission already) 2. My own local LAN Network (192.168.1.0/24) 2. My own local LAN Network
Permission secured from client?	Yes
Phases covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery Phase 3-5: In Progress **Phase 3-5:** In Progress

1. Liability Disclaimer

I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. Introduction

This report covers footprinting the networkwalks.com domain using multiple Kali Linux tools (W2-PM1) and scanning my own local network with Zenmap (W2-PM5). One module covers the footprinting phase and the other covers the scanning phase, so together they show how an attacker moves from gathering public information to mapping live hosts on a network. It is the Week 2 part of my ongoing cybersecurity program at Networkwalks.
All commands were run in Kali Linux (footprinting) and on a Windows PC with Zenmap installed (scanning). Every step below includes the exact command used, the result I observed, a screenshot as evidence, and a short note on why the finding matters from an attacker's point of view.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux & Windows	Operating systems used for reconnaissance and scanning activities
whatweb	Fingerprint web technologies (server, CMS, plugins, IP, headers).
nslookup	Resolve the domain name to its IP address using DNS.
curl -I	Read the HTTP response headers and exposed endpoints of the website.
wafw00f	Detect whether a Web Application Firewall protects the site.
dnsrecon	Enumerate all DNS records (SOA, NS, MX, A, SPF, TXT, SRV).
Maltego Desktop	Map OSINT entity relationships (domain, email addresses, web links).
theHarvester	Demonstrate multi-source public OSINT harvesting workflows.
Zenmap (Nmap GUI)	Scan the local subnet (192.168.1.0/24) to find live hosts, IPs and MAC addresses.

4. Activities Performed

4.1 Footprinting & Reconnaissance

I performed reconnaissance against the networkwalks.com domain using Kali Linux tools: WhatWeb, Nslookup, Curl, Wafw00f, DNSRecon, Maltego, and theHarvester. Each tool was used to collect a different type of information about the target.

First, I used WhatWeb (whatweb networkwalks.com) to identify technologies used by the website. The results identified an Apache HTTP server at IP 192.232.216.135, WordPress 7.1.4, WordPress Download Manager 3.3.76, Bootstrap 7.1.4, jQuery 3.7.1, Google Tag Manager, and the contact email info@networkwalks.com.

Using Nslookup (nslookup networkwalks.com), I resolved the domain name to its IP address via my local DNS forwarder (192.168.51.2). The result identified the public server IPv4 address 192.232.216.135.

I used Curl with the -I option (curl -I https://networkwalks.com) to inspect the HTTP response headers. This confirmed HTTP/2 200 OK on Apache, revealed caching headers (x-nginx-cache: WordPress, x-endurance-cache-level: 0), showed the __wpdm_client cookie (Secure, HttpOnly), and exposed the WordPress REST API endpoint /wp-json/.

Next, I used Wafw00f (wafw00f networkwalks.com) to determine whether a Web Application Firewall was protecting the website. The result confirmed ModSecurity (SpiderLabs) WAF.

Finally, I used DNSRecon (dnsrecon -d networkwalks.com) to enumerate DNS records. The results identified HostGator name servers (ns6135.hostgator.com at 50.87.144.87 and ns6136.hostgator.com at 192.232.216.131), BIND version "9.16.23-RH", mail server mail.networkwalks.com (192.232.216.135), TXT/SPF records, missing DNSSEC, and 8 cPanel autodiscover SRV records (_autodiscover._tcp.networkwalks.com on port 443).

Additionally, I used Maltego Desktop to visually graph the entity relationships between networkwalks.com and info@networkwalks.com, and executed theHarvester (theHarvester -d microsoft.com -l 50 -b all) to test multi-source OSINT harvesting workflows.

4.2 Network Scanning with Zenmap

For the second activity, I used Zenmap (Nmap 7.993) to perform network discovery on my local network (192.168.1.0/24). The practical required me to identify my local subnet, discover live hosts, identify their IP and MAC addresses, and generate a network topology.

I entered my local subnet 192.168.1.0/24 into Zenmap and selected Ping Scan (nmap -sn 192.168.1.0/24) to identify active hosts.

My scan scanned 256 IP addresses in 7.66 seconds and identified 7 live hosts on my network:

- 192.168.1.1 — MAC: 78:8C:B5:2A:EE:C2 (TP-Link Systems – Gateway/Router)
- 192.168.1.101 — MAC: 60:A4:B7:81:DF:65 (TP-Link Systems)
- 192.168.1.102 — Local Scanning Host (localhost)
- 192.168.1.109 — MAC: 76:E8:B7:ED:44:BC (Unknown / Randomized MAC)

192.168.1.111 — MAC: 86:29:3F:D6:F2:FC (Unknown / Randomized MAC)

192.168.1.122 — MAC: 5A:15:4D:EB:2E:9F (Unknown / Randomized MAC)

192.168.1.124 — MAC: 7E:F9:74:40:0F:27 (Unknown / Randomized MAC)

The scan results included MAC addresses and vendor details for all neighboring live hosts on the subnet.

After completing the scan, I opened the Topology section in Zenmap (Fisheye view) to visualize the network topology connecting localhost (192.168.1.102) to the six active hosts.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology information exposed	WhatWeb identified WordPress 7.1.4, WP Download Manager 3.3.76, Bootstrap 7.1.4, and jQuery 3.7.1	Attackers may use exposed technology/version information to identify software requiring further security review	● **Medium**
2	Server IP address identifiable	Nslookup resolved the domain to 192.232.216.135	Provides information about the network location of the web service	● **Low**
3	HTTP technical information exposed	Curl returned Apache/cache headers and exposed /wp-json/	May assist technology fingerprinting and further enumeration	● **Low**
4	WAF technology identifiable	Wafw00f identified ModSecurity (SpiderLabs)	Reveals information about the web application's security architecture	● **Low**
5	DNS infrastructure & daemon version exposed	DNSRecon identified BIND 9.16.23-RH, HostGator NS/mail records, 8 SRV records, and missing DNSSEC	DNS version and service records can help build a broader infrastructure profile	● **Medium**
6	Multiple live hosts visible on local network	Zenmap identified 7 live hosts on 192.168.1.0/24 (including 4 unknown/randomized MACs)	Unknown or unauthorized devices may potentially be present on a network	● **Medium**

Risk level key: ● Critical ● Medium ● Low

The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities.

The practical exercises primarily involved information gathering and host discovery. No exploitation or vulnerability validation was performed as part of these two modules.

Therefore, the presence of information such as a software version, IP address or DNS record does not by itself mean that the system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability.

6. Recommendations

Based on the observations from these activities, I recommend the following security improvements:

1. **Review publicly exposed technology information**
   Organizations should regularly review what information about their web technologies, CMS and plugins is publicly visible.
2. **Keep software updated**
   CMS platforms, plugins and other web technologies should be regularly updated and reviewed against current security advisories.
3. **Review HTTP headers**
   HTTP response headers should be reviewed to determine whether unnecessary technical information is being exposed.
4. **Review DNS records regularly**
   DNS records should be checked periodically to ensure that only required information and services are publicly exposed.
5. **Properly configure and monitor the WAF**
   Keep the WAF (ModSecurity) enabled and tuned, since it already blocks naive attacks.
6. **Perform regular internal network discovery**
   Organizations should periodically scan their own networks to identify active devices.
7. **Investigate unknown devices**
   Any unexpected device discovered during network scanning should be investigated and verified.
8. **Maintain network documentation**
   Network topology and device information should be documented and updated regularly.
9. **Perform security testing with authorization**
   Reconnaissance and scanning should only be performed against systems and networks where appropriate authorization has been provided.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking program, I completed practical activities covering footprinting, reconnaissance and network scanning.

In the footprinting activity, I used Kali Linux tools (WhatWeb, Nslookup, Curl, Wafw00f, DNSRecon, Maltego, and theHarvester) to collect information about the target domain. I learned how WhatWeb identifies web technologies, Nslookup resolves domain names, Curl inspects HTTP headers, Wafw00f identifies a WAF, DNSRecon enumerates DNS/SRV records, and Maltego/theHarvester gather public OSINT.

In the network scanning activity, I used Zenmap to scan my local subnet (192.168.1.0/24) and discover 7 active hosts. I also collected IP and MAC address information and generated a network topology.

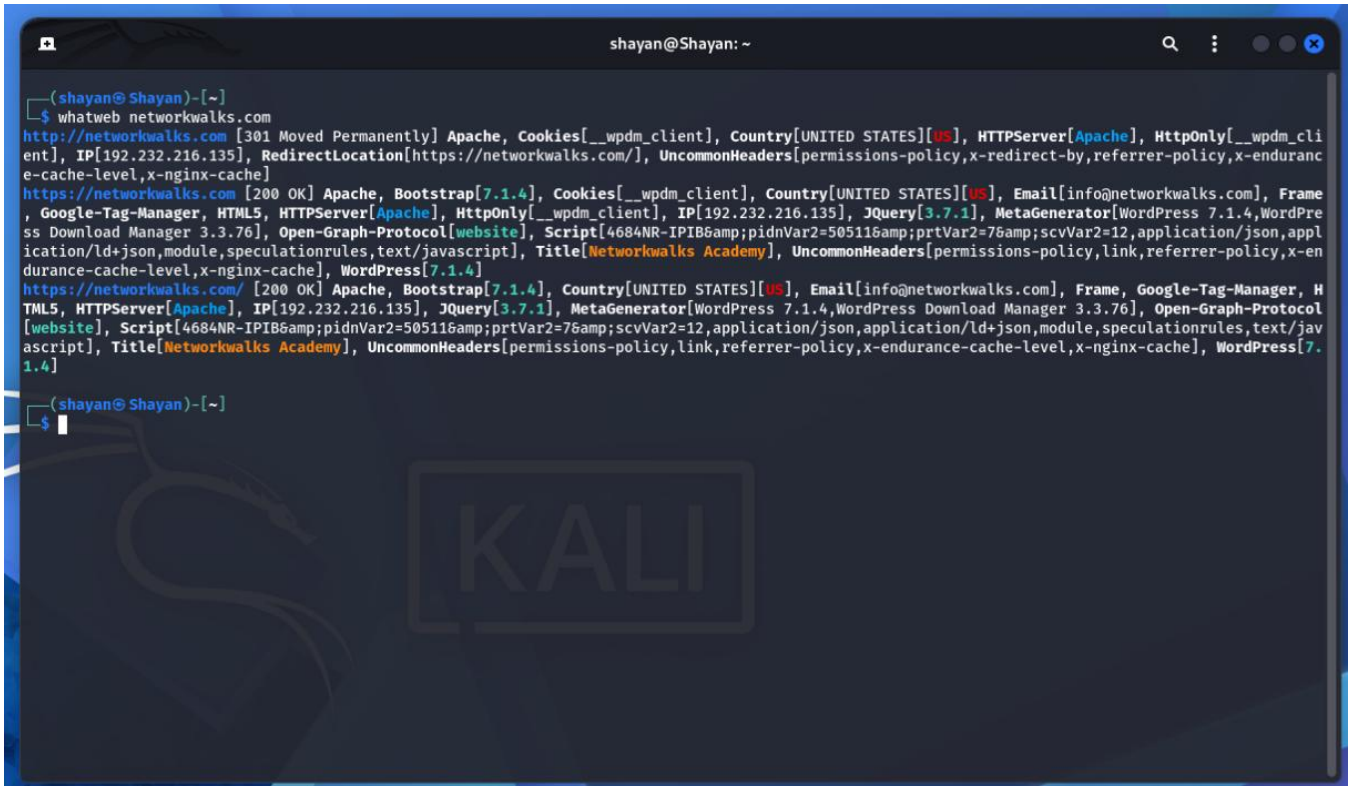
The exercises showed me that information gathering is an important part of cybersecurity. Even before attempting to exploit a system, a security professional can learn a significant amount about an environment by carefully analyzing publicly available information and network responses.

I also learned that technical findings should be documented clearly. A good cybersecurity report should explain what was performed, what was discovered, what the observation means, what risk it may create, and what can be done to reduce that risk.

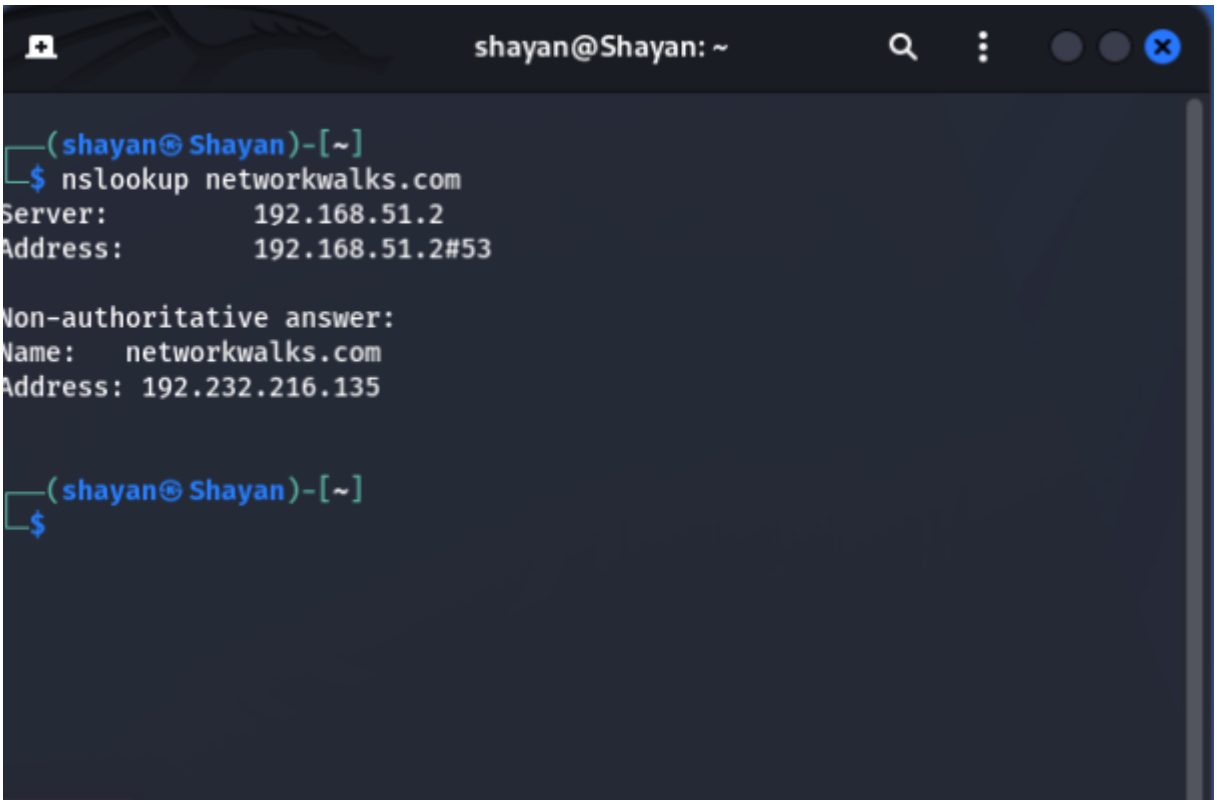
Finally, I learned that reconnaissance and scanning must always be performed within an authorized scope. These activities were completed as part of the assigned educational cybersecurity lab.

8. Evidences Collected

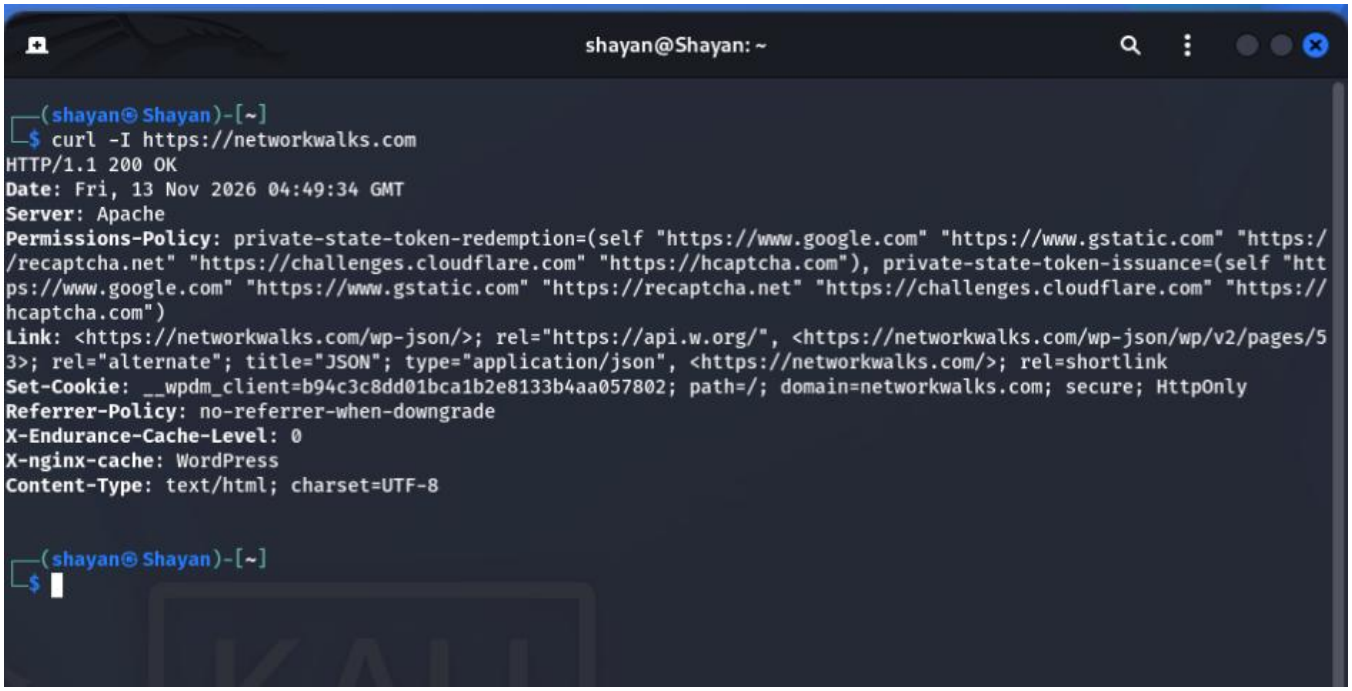
Evidence 1: WhatWeb Fingerprinting (whatweb networkwalks.com)



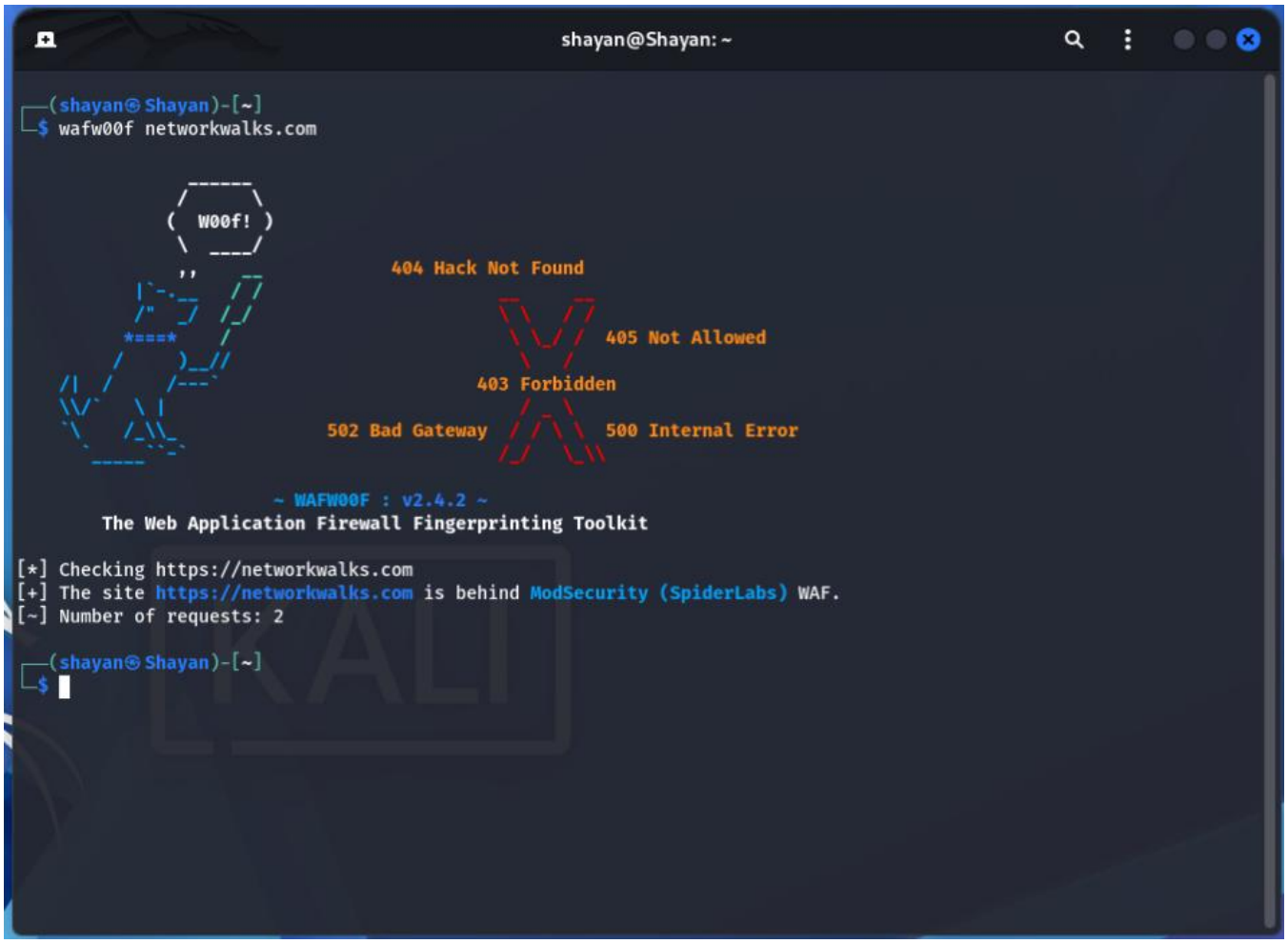
Evidence 2: DNS Resolution (nslookup networkwalks.com)



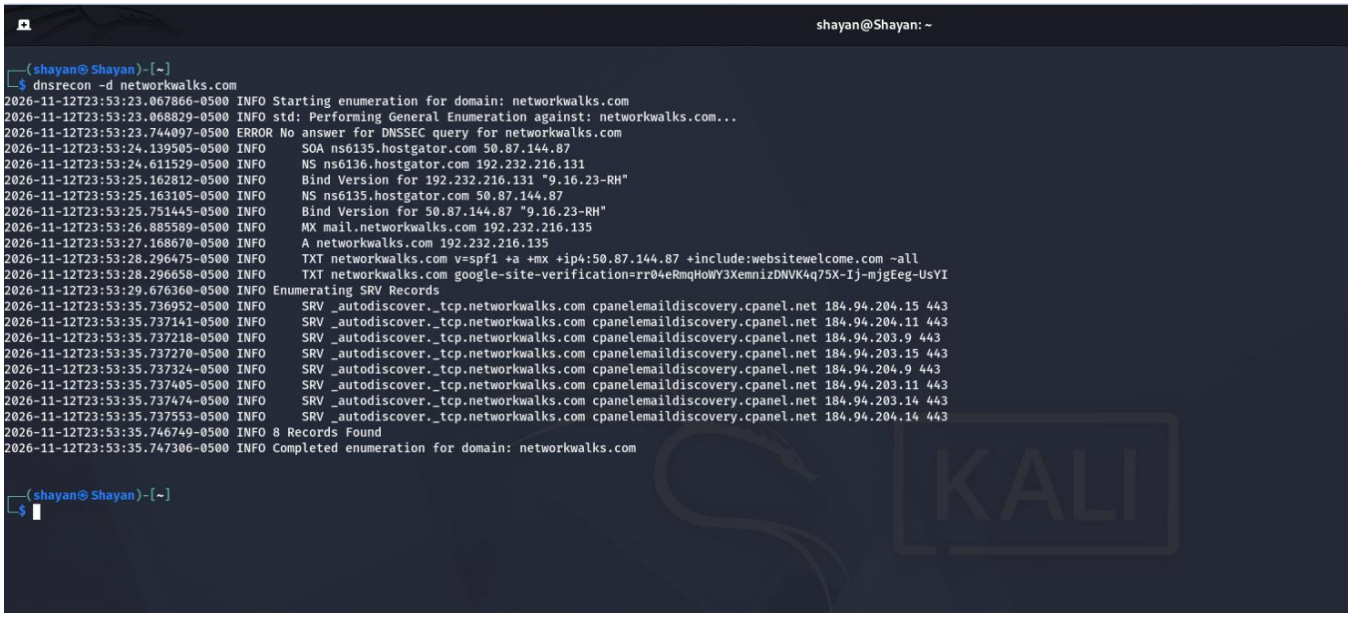
Evidence 3: HTTP Response Headers (curl -I https://networkwalks.com)



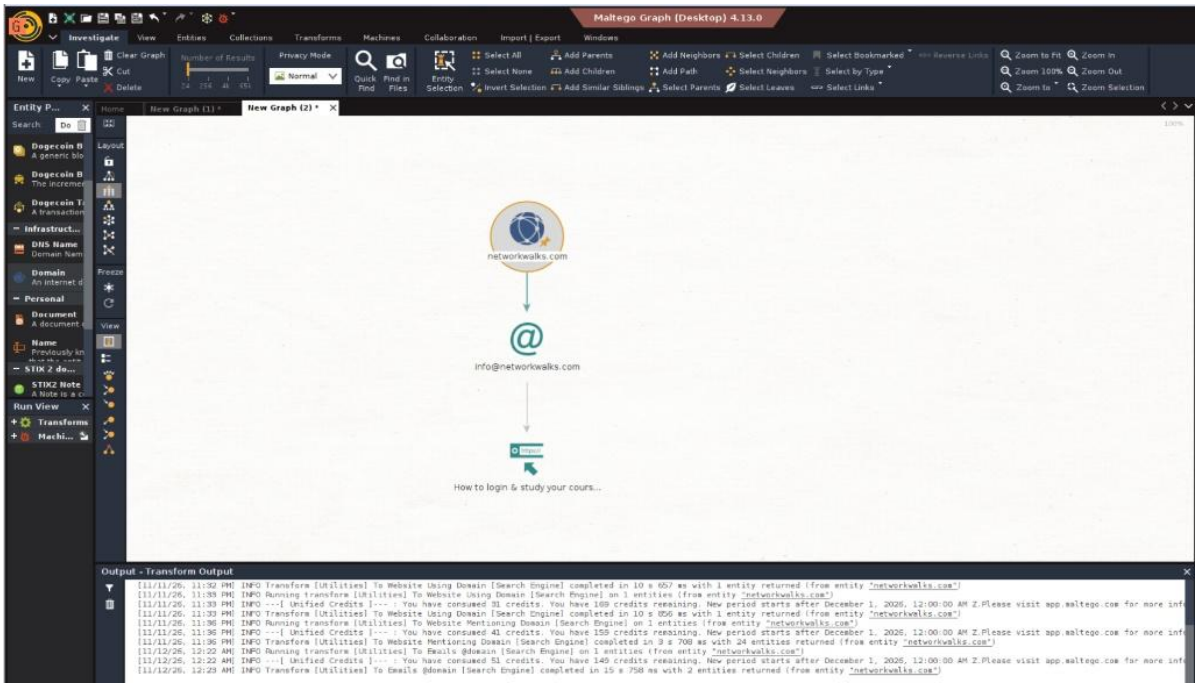
Evidence 4: WAF Detection (wafw00f networkwalks.com)



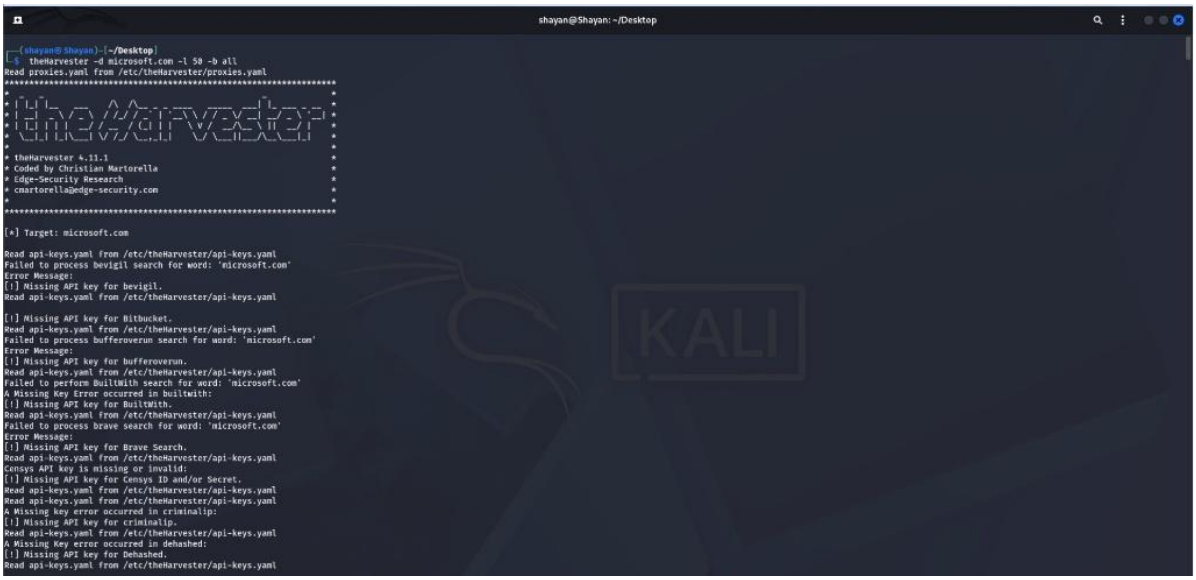
Evidence 5: DNS Record Enumeration (dnsrecon -d networkwalks.com)



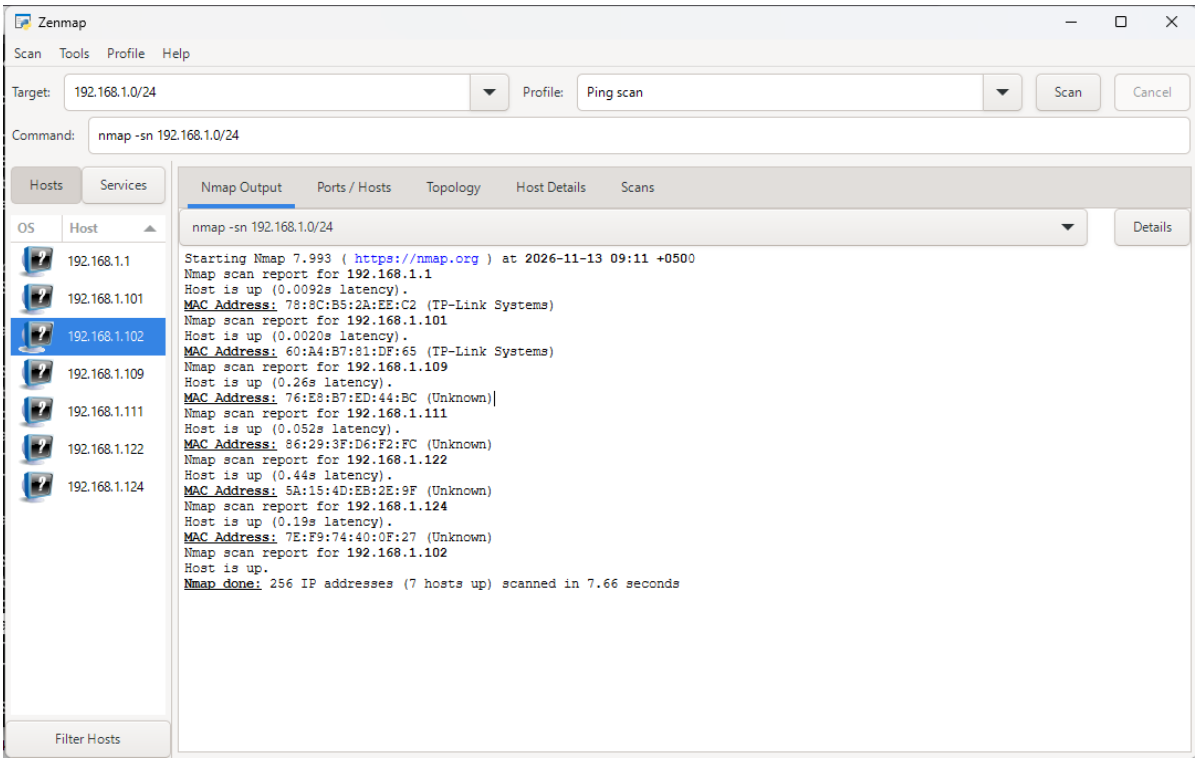
Evidence 6: Visual OSINT Entity Graph (Maltego Desktop)



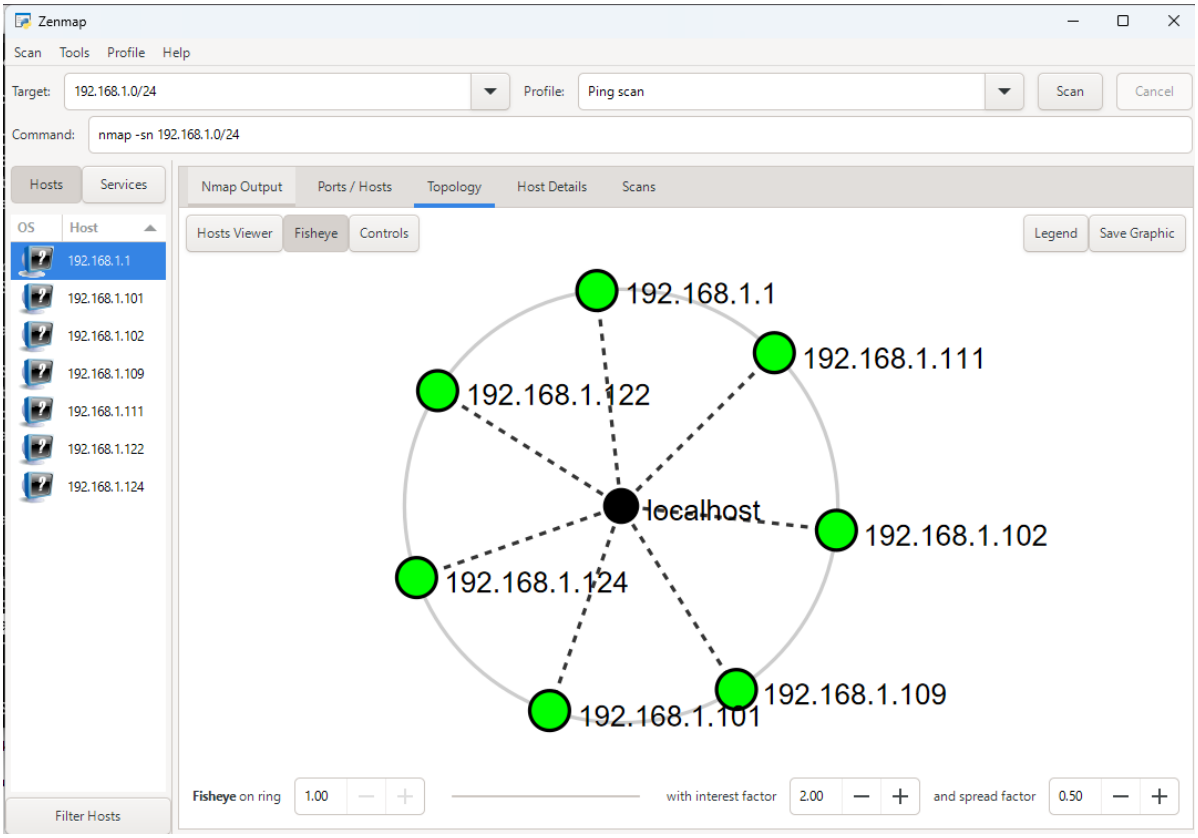
Evidence 7: Automated OSINT Workflow (theHarvester)



Evidence 8: Local Subnet Ping Scan in Zenmap (nmap -sn 192.168.1.0/24)



Evidence 9: Local Network Topology Visualization in Zenmap



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📌 **Project Information**
Program Name: Cybersecurity program at Networkwalks | **Week:** 02 | **Repository:** GitHub